Summary in Graph

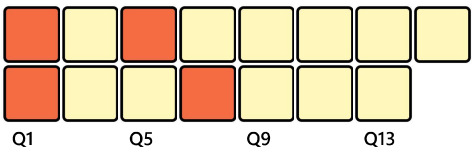
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 5)

Qs. Attempted:	8 8 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	1.33 1.33 + 0
Incorrect Attempts:	8 8 + 0	Resultant Marks:	-1.33 -1.33 + 0

Total Questions:	15 10 + 5
Total Marks:	20 10 + 10
Exam Duration:	45 Minutes
Time Taken:	45 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 370 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Given a binary attribute A that splits a set of examples E into two nonempty subsets E_1 and E_2, where:
- E_1 has $p_1 > 0$ positive examples and $n_1 > 0$ negative examples.
- E_2 has $p_2 > 0$ positive examples and $n_2 > 0$ negative examples.

The binary class label C represents whether an example belongs to the positive or negative class. Is it possible for the information gain $I(C; A) = 0$?

A. Yes, because if $p_1 = n_1 = p_2 = n_2$, $H(C \mid A) = H(C)$, and $I(C; A) = 0$.

B. No, because information gain cannot be zero when examples are split into two nonempty subsets.
C. Yes, but only when $p_1 > p_2$ and $n_1 > n_2$.
D. No, because splitting into subsets always increases information gain.

Your Answer: D | Correct Answer: A | Incorrect | Time taken: 07min 19sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose we are learning a decision tree from a dataset with M binary-valued inputs and R training points. What is the maximum possible number of leaves in the decision tree?

A. $\min(R, M)$
B. $\min(R, \log_2(M))$
C. $\max(R, 2^M)$
D. $\min(R, 2^M)$

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 05min 51sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Jonathan just trained a decision tree for a digit recognition. He notices an extremely low training error, but an abnormally large test error. He also notices that an SVM with a linear kernel performs much better than his tree. What could be the cause of his problem?

A. Decision tree is too deep
B. Decision tree is overfitting
C. Learning rate too high
D. There is too much training data

Your Answer: B;C | Correct Answer: A;B | Incorrect | Time taken: 00min 58sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following statements is/are TRUE about decision trees with different depths?

A. A tree with depth of 3 has higher variance than a tree with depth 1.
B. A tree with depth of 3 has higher bias than a tree with depth 1.
C. A tree with depth of 3 never has higher training error than a tree with depth 1.
D. A tree with depth of 3 never has higher test error than a tree with depth 1.

Your Answer: A;B | Correct Answer: A;C | Incorrect | Time taken: 02min 53sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

As part of building a decision tree, we want to measure the information gain from asking a question about a binary split on feature X_1 of the following 4 samples:

X_1	y
T	0
F	1
F	1

X_1	y
F	0

Which of the following measures the information gain of splitting the sample on the value of X_1 ?

A. $-\left(\frac{2}{3}\right)\log_2\left(\frac{2}{3}\right)+\left(\frac{1}{2}\right)\log_2\left(\frac{1}{2}\right)$
B. $-\log_2\left(\frac{1}{2}\right)+\frac{1}{4}\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right)+\frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)+\frac{3}{4}\left(\frac{2}{3}+\log_2\left(\frac{2}{3}\right)+\frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)$
C. $-\log_2\left(\frac{1}{2}\right)+\frac{3}{4}\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right)+\frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)$
D. $-\log_2\left(\frac{1}{2}\right)+\frac{1}{4}\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right)+\frac{1}{3}\log_2\left(\frac{1}{3}\right)\right)$

Your Answer: D | Correct Answer: C | Incorrect | Time taken: 09min 38sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider T_1, a decision stump (tree of depth 2) and T_2, a decision tree that is grown till a maximum depth of 4 . Which of the following is/are correct?

A. $\text{Bias}(T_1) < \text{Bias}(T_2)$
B. Variance $(T_1) < \text{Variance}(T_2)$
C. $\text{Bias}(T_1) > \text{Bias}(T_2)$
D. $\text{Variance}(T_1) > \text{Variance}(T_2)$

Your Answer: A;B | Correct Answer: B;C | Incorrect | Time taken: 04min 11sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Select the true statements about decision trees.

A. Decision trees are grown from the bottom up, meaning that we start at the leaves and repeatedly fuse them until we reach the root node.
B. Pruning improves both training and validation accuracy.
C. They can guarantee 100% training accuracy on any training set where no two identical points have different labels.
D. Decision trees are used for classification only, not regression.

Your Answer: B;C | Correct Answer: C | Incorrect | Time taken: 00min 54sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider training a decision tree given a design matrix $X=\begin{bmatrix}6 & 3\\ 2 & 7\\ 9 & 6\\ 4 & 2\end{bmatrix}$ and labels $y=\begin{bmatrix}1\\ 0\\ 1\\ 0\end{bmatrix}$. Let f_1 denote feature 1 , corresponding to the first column of X, and let f_2 denote feature 2 , corresponding to the second column. Which of the following splits at the root node gives the highest information gain? (Select one.)

A. $f_1 > 2$
B. $f_2 > 3$
C. $f_1 > 4$
D. $f_2 > 6$

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 06min 21sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Suppose you have a fixed dataset and want to train a decision tree for two-class classification. If you increase the maximum depth of the decision tree, which of the following are possible effects?

A. The test accuracy goes up.
B. The training accuracy goes down.
C. The number of pure leaves is reduced.
D. The time to classify a test point increases.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 07min 01sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following are true of decision trees? Assume splits are binary and are done so as to maximize the information gain.

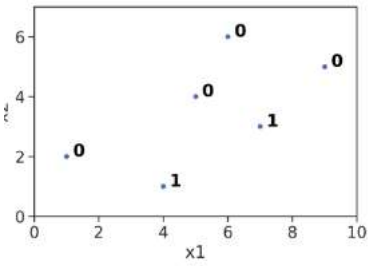
A. If there are at least two classes at a given node, there exists a split such that information gain is strictly positive.
B. As you go down any path from the root to a leaf, the information gain at each level is non-increasing
C. The deeper the decision tree is, the more likely information gain is strictly positive
D. The information gain is always strictly positive at each split in the tree.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose we are trying to train a decision tree model for a binary classification task. We denote the two classes as $\mathbf{0}$ (the negative class) and $\mathbf{1}$ (the positive class) respectively. Our input data consists of 6 sample points and 2 features x_1 and x_2.
The data is given in the table below, and is also plotted for your convenience on the right.

x_1	x_2	Label(y)
1	2	0
4	1	1
5	4	0
6	6	0
7	3	1
9	5	0



Suppose we split the root note with a rule of the form $x_i \geq \beta$, where i could be either 1 or 2 . Which of the following rules minimizes the weighted entropy of the two resulting child nodes?

A. $x_1 \geq 3$

B. $x_1 \geq 4.5$
C. $x_2 \geq 3.5$
D. $x_2 \geq 4.5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a training set with n items. Each item is represented by a d-dimensional feature vector $(x_1, \ldots, x_d)$ and a class label y. Each feature dimension is continuous (i.e., it is a real number). To build a decision tree, one may ask questions of the form "Is $x_1 \geq \theta$?" where θ is a threshold value.
Ideally, how many different θ values should we consider for the first dimension x_1 ?

A. Infinite, because x_1 is continuous
B. nd, all possible numbers in the training set
C. One, at the middle of x_1 's range
D. $n + 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider a small XOR dataset:-

x_1	x_2	y
+1	+1	0
+1	-1	1
-1	-1	0
-1	+1	1

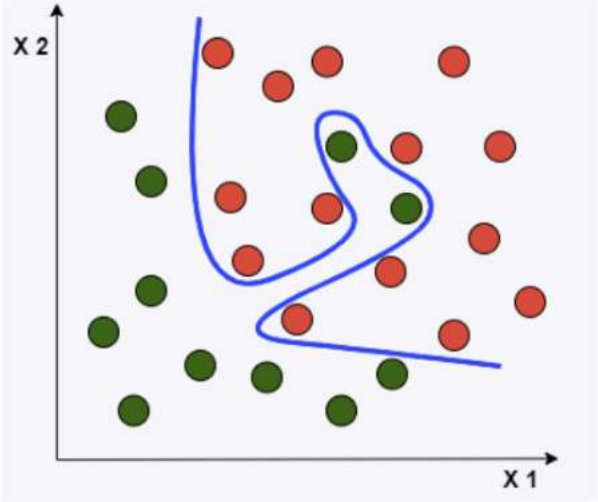
Which of the following models can achieve zero training error?

A. A decision tree with depth 1.
B. A decision tree with depth 2.
C. A perceptron.
D. Linear soft-margin SVM.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Which of the following is the most correct description for the following image, which shows the decision boundary for a classification model on a training set with two features x1 and x2 ?



A. It doesn't represent the decision boundary for a decision tree.
B. The model is likely to have a high accuracy on the test set.
C. This illustrates an example of underfitting.
D. If the model is currently regularized, to improve the predictive performance of this model, we should decrease the regularization parameter.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Suppose we are deciding how to split a particular node in a decision tree. Currently, 20 training examples (10 positive and 10 negative) belong to this node. If we split on a certain binary-valued attribute A, some examples will go down the left branch and others will go down the right branch.
Which of the following statements about the information gain for the split is/are correct?

A. The minimum possible information gain is 0 , which occurs when both branches have equal distributions of positive and negative examples (e.g., 5 positive/5 negative in both branches).
B. The maximum possible information gain is 1 , which occurs when all positive examples go down one branch and all negative examples go down the other branch.
C. The minimum possible information gain is -1 .
D. The maximum possible information gain is 0.5 .

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

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